

HCMIU Map Collaborative

A Multi-Model Database-Driven Campus Map
with Real-Time Collaboration

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Outline

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- 2 System Architecture
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Problem Context

Challenge: Campus information systems must handle

- **High schema variability** — rooms, notes, comments, trials, moderation records evolve over time
- **High relationship density** — references, follows, replies, dispute participants form a dense graph

Traditional approach: Polyglot persistence (separate document store + graph DB) $\Rightarrow$ operational overhead, synchronisation burden [1, 2]

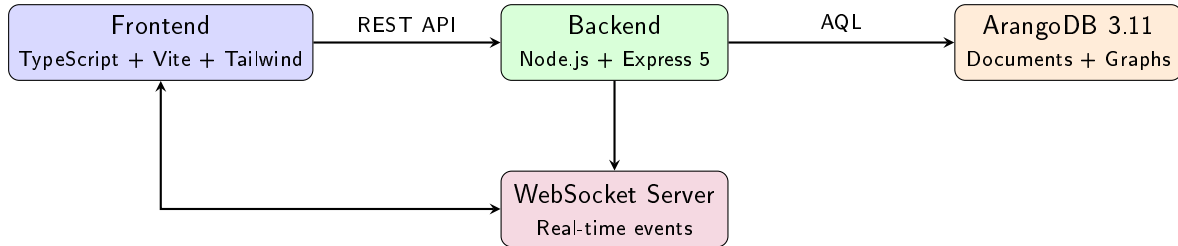
Our Approach

Use **ArangoDB**—a single multi-model engine that natively supports documents, graphs, and key-value workloads—to reduce impedance mismatch while keeping full graph query power.

Project Objectives

- ➊ **Interactive campus navigation** — 7-floor indoor map with BFS shortest-path and Traveling Salesman (Held–Karp) multi-stop optimiser
- ➋ **Collaborative knowledge graph** — entity-based discussions with cross-references, follows, notifications, and an activity feed
- ➌ **Transparent dispute resolution** — a structured “Court of Justice” trial system with judge negotiation and voting
- ➍ **Graph-native research tools** — reference discovery, full-text search, degree-of-separation queries
- ➎ **Real-time experience** — WebSocket event propagation for live updates

System Architecture



- **Deployment:** Docker Compose (3 services: DB, backend, frontend)
- **Frontend:** Hyperscript DOM library + Tailwind CSS + Anime.js
- **Auth:** Client-side Argon2i (libsodium) → server SHA-256

Multi-Model Database Design

Document Collections:

- `users` — credentials, salts, tokens
- `entities` — posts, comments, map locations, trial entities
- `follows` — user → entity subscriptions
- `notifications` — per-user alerts
- `trials` — court cases with status, judges, votes

Edge Collection:

- `entity_references` — directed edges connecting entities that reference each other

Why Multi-Model?

- Document storage for flexible schema evolution
- Graph edges for relationship traversal
- Single AQL query language for both
- No cross-system synchronisation needed

Representative AQL Queries

1. Reference Discovery (Graph Traversal):

```
FOR v, e IN 1..1 OUTBOUND @entityId entity_references
  RETURN { entity: v, edge: e }
```

2. Degree of Separation (Shortest Path):

```
FOR v IN OUTBOUND SHORTEST_PATH
  @fromId TO @toId entity_references
  RETURN v
```

3. Full-Text Search:

```
FOR e IN entities
  FILTER CONTAINS(LOWER(e.title), LOWER(@query))
    OR CONTAINS(LOWER(e.body), LOWER(@query))
  RETURN e
```

HCMIU CAMPUS INTELLIGENCE



HCMIU Map Platform

A cinematic workspace for navigation, collaboration, and multi-floor planning. Pin destinations, rehearse routes, and bring discussions into every room.

Realtime pathfinding

Collaborative graph

Research APIs

FLOORS

7

Stack-aware routing and lift coverage

GRAPH ENTITIES

Live

Rooms, users, trials, references

REALTIME

WS

WebSocket notifications & follows

TRAVERSAL

TSP

Multi-stop optimizer with floor jumps

Launchpad



View Map

Explore floors, rooms, gates, and key facilities at a glance.



Find Shortest Path

Compute the quickest route between two places across floors.

Interactive Campus Map

HCMIU MAP WORKSPACE

Campus Explorer

Browse the map, jump to any room with instant search, and open collaborative discussions inline.

Exit

MODE

Pick a spot

Click the map or use quick search to open the thread.

FLOOR

—

Use keyboard search to jump between floors.

REFERENCES

0

Entities pointing at this map location.

COMMENTS

0

Conversation attached to this room or stairs.

Floor 1

STAIRS

LIFT

A2 102

Location Spotlight

Search any room, lab, gate, or lift to center the map and instantly open its discussion thread.

Search...

Shortest Path (BFS)

- Breadth-first search on grid graph
- Handles **inter-floor routing** via stairs and lifts
- Visual path overlay on floor maps
- Time complexity: $O(V + E)$

User Flow:

Select start → Select end →
View floor-by-floor directions

Traveling Salesman (Held–Karp)

- Bitmask dynamic programming
- Optimal visit order for n destinations
- Multi-floor route planning
- Time complexity: $O(2^n \cdot n^2)$

User Flow:

Add locations → Solve →
View optimised route with total distance

Collaborative Knowledge Graph

Exit

Entities: 1316

Trials: 0

Notifications: 0

Mode: Public

Hub

Auth

Entities

Trials

Research

Notifications

Activity

Tutorial

Entities

Floor 7: LIFT (A1, right) map_location

Collaborative thread for Floor 7: LIFT (A1, right)

id: entity_map_f0280765fa47f7a91f3e1700

refs: none

Floor 7: LIFT (A1, left) map_location

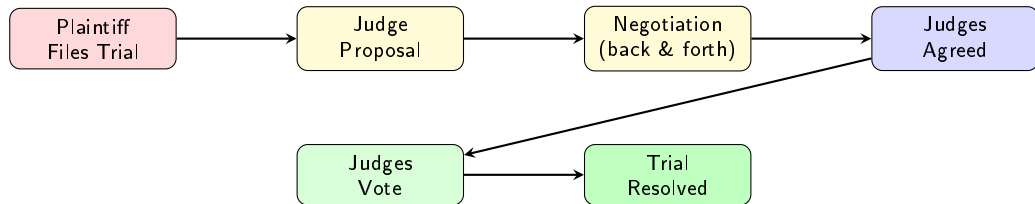
Collaborative thread for Floor 7: LIFT (A1, left)

id: entity_map_8b611b17a515b9173fb6ddaa

refs: none

Floor 7: LIFT (A2, right) map_location

Court of Justice — Trial System



- Plaintiff files trial against defendant over a disputed entity
- Both parties negotiate judge selection (proposal ↔ counter-proposal)
- Agreed judges cast votes: plaintiff / defendant / no_winner
- Comments tagged by role: [plaintiff], [defendant], [judge], [spectator]

Reference Discovery

- Query entities by ID
- View all outbound references
- Explore the citation network

Full-Text Search

- Search across all entity titles and bodies
- Keyword-based discovery
- Cross-entity relationship finding

Degree of Separation

- Find shortest reference path between any two entities
- Uses ArangoDB's native `SHORTEST_PATH` function
- Visualises intermediate hops

Mention Discovery

- Find which entities mention a given entity
- Reverse reference traversal
- Map cross-entity relationships

Real-Time Updates via WebSocket

Event Types:

- `entity.created` — new post or comment
- `entity.updated` — edit to existing entity
- `entity.deleted` — entity removed
- `trial.created` — new trial filed
- `trial.updated` — vote or status change
- `notification.created` — user alert

Result: When User A posts a comment, User B sees it instantly without page refresh—verified in multi-user E2E tests.

Design Decisions

- RFC 6455 WebSocket protocol
- Token-authenticated connections
- Server-push model (no polling)
- All connected clients auto-refresh

Technology Stack & Testing

Technology Stack:

Layer	Technology
Frontend UI	TypeScript, Vite, Tailwind Hyperscript, Anime.js
Backend	Node.js, Express 5
Database	ArangoDB 3.11
Real-time	WebSocket (ws)
Auth	libsodium (Argon2i)
Deploy	Docker Compose

Testing & Validation:

- **API unit tests** — all REST endpoints
- **E2E tests** — full Docker Compose stack
- **Backend stress benchmark** (25,461 requests, 0 failures)
 - Health check: 3,250.83 req/s, p95 11.82 ms
 - Full-text search: 579.04 req/s, p95 20.97 ms
- **8 demo videos** — Puppeteer-automated screen recordings covering:
 - Map navigation
 - Multi-user real-time
 - Trial system workflow
 - Research tools (4 demos)
 - General platform overview

Conclusions & Future Work

Key Contributions:

- 1 Demonstrated that a **multi-model DBMS** (ArangoDB) can effectively serve both document storage and graph traversal needs in a single engine
- 2 Built a **complete campus platform** combining wayfinding, collaboration, governance, and research
- 3 Implemented **graph-native features** (reference discovery, degree-of-separation, mention traversal) that would be complex in a document-only or relational system
- 4 Validated correctness through **comprehensive automated testing**

Limitations & Future Work:

- Access control is creator-based; role-based permissions needed
- Event ordering relies on eventual consistency; formal guarantees desirable
- Map data is hardcoded; a map editor would improve extensibility
- Extend benchmarking to sustained mixed-write workloads and larger datasets

References

-  M. Stonebraker and U. Çetintemel, “One Size Fits All: An Idea Whose Time Has Come and Gone,” *Proc. ICDE*, pp. 2–11, 2005.
-  P. Sadalage and M. Fowler, *NoSQL Distilled: A Brief Guide to the Emerging World of Polyglot Persistence*, Addison-Wesley, 2012.
-  ArangoDB Inc., “ArangoDB—the Multi-Model NoSQL Database,” <https://www.arangodb.com/>, 2024.
-  A. Zimmerman and M. Martin, “Post-occupancy evaluation: benefits and barriers,” *Building Research & Information*, vol. 29, no. 2, pp. 168–174, 2001.

Thank You

Questions?

Live Demo: <https://hcmiumap.huynhtrankhanh.com/>

Source Code: <https://github.com/huynhtrankhanh/hcmiu-map-collaborative>